Control Sensitivity and the Cost of Certifying Small Policy Improvements

Kyunghyun Lee [a], Yeoneung Kim [b,⋆]

[a] *Department of Computer Science, Gachon University, 1342 Seongnam-daero, Sujeong-gu, Seongnam, Gyeonggi-do 13120, Republic of Korea*

[b] *Department of Industrial Engineering, Yonsei University, 50 Yonsei-ro, Seodaemun-gu, Seoul 03722, Republic of Korea*

Abstract

Certifying a small controller improvement can require many simulation trajectories, even when common random numbers reduce comparison noise. We ask when control sensitivity prevents this growth, counting the simulation needed to certify that sensitivity. Baseline closed-loop sensitivity and an enforced action displacement d give a pathwise paired-return support of order d, even for discontinuous updates. For gains proportional to d, this support permits a displacement-independent validation budget; rules uniformly valid over an unrestricted fixed ordinary range face an $\Omega(d^{-1})$ barrier on the same favorable law. We acquire the support from replicated Gaussian transitions with unknown noise scale. A finite-power small-gain margin gives a uniform state envelope and an explicit calibration budget independent of d, allowing transient amplification and matrix norms above one. Every Schur-stable linear baseline admits such a certificate. Regional sensitivity can lose the saving through the cost of estimating escape probability. On 24 controlled plants, shared disturbances reduce comparison variance by a mean factor of 1,796, yet ordinary-range testing makes no adoptions at 512 pairs; supplying paired support to the same test certifies all 24.

Key words: Stochastic control; Policy improvement; Incremental sensitivity; Common random numbers; Sample complexity; Statistical certification.

1 Introduction

A controller that already performs reasonably well may need only a small correction. Verifying that correction can nevertheless consume many simulation trajectories: its return gain is small, and accepting a non-improving controller must remain unlikely. The question is whether reliable verification must become more expensive as the permitted correction shrinks. We study when control sensitivity prevents this increase, after charging the simulation needed to certify the sensitivity itself.

Our answer uses simulations in which the two controllers share initial states and disturbances. Baseline closed-loop sensitivity can make the range of their return difference shrink with the correction, so that the signal and its worst-case uncertainty shrink together. A finite-power certificate makes this information available beyond one-step contraction. With a fixed robust margin and a gain proportional to the correction size, acquiring the certificate and validating the gain both require budgets independent of that size. We count both stages in the cost.

Such coupling can make the return difference much less variable than either return separately. Common random numbers (CRN) are an established simulation technique, also used in policy search (Glasserman and Yao, 1992; Ng and Jordan, 2000; Strens and Moore, 2002). Pathwise Lipschitz regularity can already imply quadratic decay of comparison variance as a parameter change shrinks (Glasserman and Yao, 1992, Lemma 6.1).

The distinction can be seen from a constant positive paired difference. A verifier that knows only a fixed ordinary return range must still allow a distribution with a rare large loss. Such a non-improving distribution can produce the same positive observations until that loss is encountered. A valid pathwise sensitivity bound can exclude the large loss before validation starts. Thus small variance and a small justified support provide different information. Range-dependent terms in bounded-mean estimation are well established (Mnih et al., 2008;

⋆ Corresponding author: Yeoneung Kim. Email: yeoneung@yonsei.ac.kr.

Email addresses: kyunghyunlee@gachon.ac.kr (Kyunghyun Lee), yeoneung@yonsei.ac.kr (Yeoneung Kim).

Waudby-Smith and Ramdas, 2024); we connect their consequences to the size of a controller correction.

Safe policy improvement provides performance guarantees for replacing a baseline (Pirotta et al., 2013; Thomas et al., 2015), and residual learning constructs corrections to existing controllers (Johannink et al., 2019). Lipschitz decision-process analysis bounds expected performance (Pirotta et al., 2015; Asadi et al., 2018). Incremental stability (Angeli, 2002) also informs imitation learning sample complexity (Tu et al., 2022) and the cost of learning robust stability certificates (Zhang et al., 2022). Our task is the post-training certification of a frozen candidate's return gain. Baseline sensitivity bounds each paired observation, including for a discontinuous candidate.

Data-efficient policy optimization (Zhao et al., 2025) and the connections between direct and identification-based control (Dörfler et al., 2023) concern controller construction. We study information for a subsequent adoption decision and its acquisition cost. Model-based gain certification may avoid return validation altogether; the ordinary-range barrier concerns its stated information class.

Statistical control analysis relates sample requirements to stability margins (Gatsis and Pappas, 2021), and closed-loop identification admits non-asymptotic confidence constructions (Szentpéteri and Csáji, 2023). Here replicated Gaussian transitions acquire a return envelope at a fixed matrix accuracy. Finite powers permit transient amplification and, with a robust margin, nonlinear residuals. Under fixed discount the pair budget is uniform in finite horizon; transitions per pair still grow with horizon. Regional sensitivity can lose the saving through the cost of escape information.

Section 2 states the cost comparison, Section 3 derives the control envelopes, and Section 4 accounts for acquisition. Sections 5 and 6 treat regional sensitivity and numerical evidence. The appendices give the calibration proofs and sequential power details.

2 Certification and the verifier's information

2.1 The adoption decision

Fix a baseline B, horizon H and discount $0 < \gamma \leq 1$. Let

$$R(\pi) = \sum_{t=0}^{H-1} \gamma^t r(X_t, \pi_t(X_t)), \qquad J(\pi) = \mathbb{E}R(\pi).$$

The candidate is frozen before validation. Each return pair shares its initial state and disturbances; different pairs are independent. Coupling preserves the marginal laws, so $D = R(\pi) - R(B)$ has mean $\mu = J(\pi) - J(B)$. The verifier decides whether to adopt the candidate, with false adoption meaning acceptance when $\mu \leq 0$.

The information supplied to that decision is central. If both returns lie in a common interval of width B_g, then $|D| \leq B_g$ follows from the marginal ranges. A sharper *paired support* $|D| \leq b_d$ uses the coupling and properties of the two trajectories. A small empirical variance or maximum does not itself justify that restriction.

For a justified half-width $b > 0$, write

$$\mathcal{P}_b = \{P : P([-b, b]) = 1\},$$
$$\mathcal{P}_b^0 = \{P \in \mathcal{P}_b : \mathbb{E}_P D \leq 0\}.$$

An adoption rule with stopping time τ is uniformly valid at level α if $\sup_{P \in \mathcal{P}_b^0} P(\tau < \infty) \leq \alpha$. Its power at budget N is $P(\tau \leq N)$ under a favorable law. Validity alone permits always retaining the baseline; a cost comparison therefore also specifies the probability of accepting a useful update.

2.2 Why the ordinary range can be costly

The range-only null class contains every law compatible with the stated range and a nonpositive mean. Additional plant information may exclude some of those laws. The following bound applies when the verifier must remain valid over the full class.

Proposition 1 (Range-only adoption barrier) *Let an adoption rule, possibly randomized, be uniformly valid over $\mathcal{P}_b^0$. If it adopts by N with probability at least $1 - \beta$ under $P \in \mathcal{P}_b$ with mean $\mu > 0$, and $1 - \beta > \alpha$, then*

$$N \geq \frac{\log((1-\beta)/\alpha)}{\log(1 + \mu/b)}. \tag{1}$$

This includes favorable laws with zero variance.

PROOF. Set $p = b/(b + \mu)$ and $Q = pP + (1 - p)\delta_{-b}$, so $\mathbb{E}_Q D = 0$. Generate Q observations using independent mixture indicators. With probability p^N the first N observations all come from P; conditional on this event their joint law, including independent internal randomization, is the favorable one. All earlier stopping opportunities are included. Consequently $\alpha \geq Q(\tau \leq N) \geq p^N P(\tau \leq N) \geq p^N(1 - \beta)$, which yields (1).

For the constant favorable law $P = \delta_\mu$, the null in the proof produces $+\mu$ with probability $p = b/(b+\mu)$ and $-b$ otherwise. Its harmful outcome has probability $\mu/(b + \mu)$, allowing positive records of order b/μ even at zero favorable variance. A justified variance bound below $b\mu$ could exclude this null. The result concerns power within a budget, not every realized stopping time.

2.3 How a shrinking support changes the cost

Suppose instead that $|D_i| \le Cd$ and $\mu_d \ge md$ for fixed $C, m > 0$. A fixed-sample rule adopts when

$$\overline{D}_N > Cd\sqrt{2\log(1/\alpha)/N}. \tag{2}$$

Hoeffding's inequality (Hoeffding, 1963) gives false adoption at most α and power at least $1-\beta$ whenever

$$N \ge 2(C/m)^2\big(\sqrt{\log(1/\alpha)} + \sqrt{\log(1/\beta)}\big)^2. \tag{3}$$

The null upper-tail allowance and the favorable lower-tail allowance are both proportional to $Cd/\sqrt{N}$. Comparing their sum with the gain md cancels d from the required budget. Supplying only a fixed B_g instead requires $\Omega(d^{-1})$ pairs by Proposition 1, since $\mu_d \le Cd$. The positive gain-to-displacement margin matters: support of order d alone does not give constant cost for gains $o(d)$.

The numerical study uses the established sequential betting mixture (Waudby-Smith and Ramdas, 2024)

$$M_n = \sum_j w_j \prod_{i=1}^{n}(1 + \lambda_j D_i/b), \qquad 0 \le \lambda_j \le 1, \qquad \sum_j w_j = 1. \tag{4}$$

Nonnegative weights and stakes are fixed before validation. Under the null, M_n is a nonnegative supermartingale starting at one, so adoption at $M_n \ge 1/\alpha$ is anytime-valid by Ville's inequality. An independent calibration failure allowance is added once, and testing levels are summed across multiple updates. A support violation triggers fallback to the baseline. Appendix B gives a power bound for a specified stake.

A paired return costs $2H$ simulator transitions; a replicated one-step reset costs two. The acquisition analysis counts these post-training transitions. The experiments also report totals including their fixed training budget. Fresh validation observations are separate from those used to fit the update.

3 From control sensitivity to paired support

3.1 Why baseline sensitivity is sufficient

Consider coupled dynamics $X_{t+1} = f(X_t, A_t) + W_t$, with common X_0 and W_t in both branches. In one declared state norm let $F_B(x) = f(x, B(x))$ be κ-Lipschitz, f be L_a-Lipschitz in action, B be L_B-Lipschitz, and r be separately (L_x, L_u)-Lipschitz. These bounds are global for now. Require

$$\|\pi_t(x) - B(x)\| \le d \qquad \text{for every } x, t. \tag{5}$$

No continuity or derivative of π_t is assumed. The condition concerns the *implemented* action after any safety filter, projection or clipping; a bound on a raw learned command would not be sufficient if its subsequent implementation violated the bound. It also permits time-dependent feedback over the fixed evaluation horizon.

The useful decomposition evaluates the baseline at the candidate state. It separates the candidate's local action correction from propagation through the baseline closed loop. Comparing the candidate feedback at two different states instead would introduce regularity of π_t that the argument does not need.

Theorem 2 (Pathwise support) *For finite $\kappa \ge 0$, set $E_0 = 0$ and $E_{t+1} = \kappa E_t + L_a d$. Every coupled trajectory satisfies $\|X_t^\pi - X_t^B\| \le E_t$, and*

$$|D| \le b_d := \sum_{t=0}^{H-1} \gamma^t\big[(L_x + L_u L_B)E_t + L_u d\big]. \tag{6}$$

If $r(x, \pi_t(x)) = r(x, B(x)) =: r_B(x)$ pointwise and $\mathrm{Lip}(r_B) \le L_{r_B}$, the sharper bound is

$$|D| \le b_d^{\mathrm{eq}} := L_{r_B} \sum_{t=0}^{H-1} \gamma^t E_t. \tag{7}$$

Both bounds are linear in d at fixed horizon and may be intersected with any independently justified ordinary range.

PROOF. Write $e_t = \|X_t^\pi - X_t^B\|$. Shared noise cancels. Adding and subtracting $f(X_t^\pi, B(X_t^\pi))$ gives

$$\begin{aligned} e_{t+1} &\le \|F_B(X_t^\pi) - F_B(X_t^B)\| \\ &\quad + L_a\|\pi_t(X_t^\pi) - B(X_t^\pi)\| \\ &\le \kappa e_t + L_a d. \end{aligned}$$

Since $e_0 = 0$, induction gives $e_t \le E_t$. The actions on the two branches differ by at most $d + L_B E_t$, proving (6). Under the pointwise reward identity, their stage rewards are $r_B(X_t^\pi)$ and $r_B(X_t^B)$; this proves (7).

The recursion gives $E_t = L_a d \sum_{j<t} \kappa^j$, with an empty sum at $t = 0$. Thus d sets the observation scale, while the coefficient records horizon and closed-loop amplification.

The reward identity holds, for example, for $r(x, a) = r_0(x) + c_0(x, \|a\|)$ when the update preserves the baseline action norm at each state. It removes direct effort

variation but not the first-order effect of changed state occupation.

At fixed horizon the envelope is finite even for an expanding baseline, although amplification may make it unhelpful. When $\kappa < 1$, $E_t = L_a d(1-\kappa^t)/(1-\kappa)$ is bounded uniformly in time. The return bound is then uniform in horizon for fixed $\gamma < 1$, and grows at most linearly in horizon for $\gamma = 1$. The finite-power argument below obtains the same time-uniform state control when the one-step criterion fails. State/action-dependent disturbances require the corresponding pathwise same-noise regularity; independent unbounded disturbances do not cancel.

3.2 A stochastic control example with no paired noise

Corollary 3 (A stochastic separation) *There is a horizon-two stochastic control family with bounded Lipschitz rewards, displacement d, nonzero marginal return variance and constant paired gain $\mu_d \asymp d$, for which ordinary-range certification needs $\Omega(d^{-1})$ pairs while paired-support certification needs $O(1)$. The order d in Theorem 2 cannot generally be reduced.*

PROOF. Let $x_0 = 0$, $B = (1,0)$, $\pi = (\cos\theta, \sin\theta)$, $f(x,a) = a_2$, $r(x,a) = \mathrm{clip}(x,-2,2)$, $H = 2$ and $\gamma = 1$. Use shared $W_0 \sim \mathrm{Unif}[-1,1]$ and $\theta = 2\arcsin(d/2)$, $0 < d \le 1/2$. Both action norms are one and their distance is d. With $\kappa = 0$ and $L_a = L_{r_B} = 1$,

$$R(B) = W_0, \quad R(\pi) = W_0 + \mu_d, \quad \mu_d = d\sqrt{1 - d^2/4}. \tag{8}$$

The reward remains in its linear region on these trajectories. Each return has variance $1/3$; independent evaluation gives contrast variance $2/3$, whereas CRN gives $D = \mu_d$ almost surely. The ordinary contrast bound $B_g = 4$ is independent of d. Theorem 2 gives $b_d^{\mathrm{eq}} = d$, and $\mu_d/d \to 1$, proving sharpness. Proposition 1 gives the ordinary-range lower bound. A betting component with $\lambda = 1$ and weight $w > 0$ crosses by

$$N = \left\lceil \frac{\log(1/(\alpha w))}{\log(1 + \mu_d/b)} \right\rceil.$$

For $b = d$ this is bounded independently of d; for $b = 4$ it has order d^{-1}. The observations themselves are identical.

The example makes the information restriction explicit. In the true plant, one paired observation equals the exact gain, but a range-only verifier must remain valid for the contaminating null from Proposition 1. A verifier given the model-derived support $[-d, d]$ can exclude the large harmful outcome used by that null. The separation therefore does not contradict the elementary fact that a *known* constant positive difference establishes improvement.

3.3 Matrix powers and actuator directions

A scalar norm bound treats every perturbation as if it were aligned with the most amplified direction at every step. This can be conservative for a nonnormal baseline or an underactuated system. A finite-horizon matrix calculation retains propagation by matrix powers and, when available, the directions in which a correction actually enters.

Suppose $F_B(x) = Mx + h(x)$ with $\mathrm{Lip}(h) \le \ell$, and a calibration event gives $\|M - \widehat{M}\|_2 \le \varepsilon$. Put $p_j = \|\widehat{M}^j\|_2$ and $L_\Delta = \varepsilon + \ell$.

Proposition 4 (Finite-horizon envelope) *Set $E_0 = 0$ and, for $t \ge 1$,*

$$E_t = L_a d \sum_{j=0}^{t-1} p_j + L_\Delta \sum_{s=0}^{t-1} p_{t-1-s} E_s. \tag{9}$$

The state and reward bounds of Theorem 2 hold with this envelope, without requiring $\|\widehat{M}\|_2 < 1$. If $f(x,a) - f(x,B(x)) = G_0 v(x,a)$ for known G_0 and $\|v(x,a)\| \le \|a - B(x)\|$, replace the first term by $d\sum_{j<t}\|\widehat{M}^j G_0\|_2$.

PROOF. Let $z_t = X_t^\pi - X_t^B$, $h_\Delta(x) = (M - \widehat{M})x + h(x)$ and $v_t = f(X_t^\pi, \pi_t(X_t^\pi)) - f(X_t^\pi, B(X_t^\pi))$. Then $z_0 = 0$, $\mathrm{Lip}(h_\Delta) \le L_\Delta$ and

$$z_t = \sum_{s=0}^{t-1} \widehat{M}^{t-1-s}[h_\Delta(X_s^\pi) - h_\Delta(X_s^B) + v_s].$$

Take norms and use induction for (9). If $v_s = G_0 u_s$ with $\|u_s\| \le d$, retain $\widehat{M}^{t-1-s}G_0$ in the forcing term. The stage-reward argument is unchanged. All coefficients are independent of d, so the recursion remains linear in d.

The input refinement uses the same matrix-confidence event. For a target actuator $G_0 T$ with unknown $\|T\|_2 \le 1$, its assumption holds by taking $u_s = T[\pi_s(X_s^\pi) - B(X_s^\pi)]$. It can improve an underactuated envelope because transiently amplified directions need not be directly actuated.

3.4 A uniform envelope from finite powers

A matrix can amplify a perturbation initially and still contract after several steps. A useful block must leave enough of that contraction to absorb matrix uncertainty and the nonlinear residual. For a fixed search cap K,

define

$$\widehat{S}_k = \sum_{j=0}^{k-1} p_j, \qquad \widehat{\eta}_k = 1 - p_k - L_\Delta \widehat{S}_k,$$
$$\widehat{\Gamma} = \min_{1 \le k \le K : \widehat{\eta}_k > 0} \widehat{S}_k / \widehat{\eta}_k. \tag{10}$$

An empty minimum means that no certificate is available. The ratio balances amplification within a block against the remaining contraction margin. The proof uses geometric series and the classical small-gain argument (Doyle et al., 1992). The resulting return support can be certified from data, as developed in Section 4.

Corollary 5 (Uniform envelope) *On the event in Proposition 4, a finite $\widehat{\Gamma}$ gives, for all $t \ge 0$,*

$$\|X_t^\pi - X_t^B\| \le L_a d \widehat{\Gamma}. \tag{11}$$

Writing $H_\gamma = \sum_{t=0}^{H-1} \gamma^t$, a valid support is

$$b_H = d H_\gamma [L_u + (L_x + L_u L_B) L_a \widehat{\Gamma}]. \tag{12}$$

Under the pointwise reward identity use

$$b_H = d H_\gamma L_{r_B} L_a \widehat{\Gamma}.$$

For known actuator image G_0, the numerator $L_a \widehat{S}_k$ in the state bound may be replaced by $\sum_{j<k} \|\widehat{M}^j G_0\|_2$.

PROOF. For a feasible k, $p_k < 1$ and grouping powers into blocks gives

$$\sum_{j \ge 0} p_j \le \widehat{S}_k / (1 - p_k).$$

For finite T, set $V_T = \max_{t \le T} \|X_t^\pi - X_t^B\|$. Proposition 4 yields

$$V_T \le \frac{\widehat{S}_k}{1 - p_k} (L_a d + L_\Delta V_T).$$

Rearranging and minimizing over feasible blocks proves (11); the reward bounds follow as before. Grouping the actuator products in the same way gives the refined numerator, with the same denominator $\widehat{\eta}_k$.

All block lengths use one matrix-confidence event, so minimizing the bound needs no extra failure allocation. The certificate is reusable across horizons and displacements; separate adoption decisions need their testing allocations. For $\gamma < 1$, $H_\gamma \le (1 - \gamma)^{-1}$ gives a horizon-uniform coefficient. At fixed horizon, linearity in d already follows from Proposition 4 without stability.

4 Acquiring the envelope and accounting for cost

The control envelope changes the return-validation cost only if it can be justified at an acceptable cost of its own. We give a concrete calibration scheme using a simulator that can reset to chosen states. At each reset it produces two baseline transitions with independent noise. Their difference estimates noise scale; their average estimates the baseline response. These calibration replicates are separate from the common-noise trajectory pairs used for validation.

4.1 Certifying baseline sensitivity

Assume $F_B(x) = Mx + h(x)$, $h(0) = 0$ and a known global $\mathrm{Lip}(h) \le \ell$. At independent reset states $x_i \in \mathbb{R}^n$, observe

$$y_{ij} = F_B(x_i) + w_{ij}, \qquad j = 1, 2,$$

where the noises are independent $N(0, \sigma^2 I_n)$. Let X have rows $x_i^\top$, Y_j have rows $y_{ij}^\top$, and let $G = X^\top X$ be nonsingular. Compute

$$\widehat{M}^\top = G^{-1} X^\top (Y_1 + Y_2)/2, \qquad s^2 = \frac{\|Y_1 - Y_2\|_F^2}{2nN}.$$

The replicate difference cancels the unknown response, including the nonlinear residual. Gaussianity makes this difference independent of the replicate mean. It therefore supplies the noise estimate needed to calibrate regression uncertainty. Neither M, h nor σ is supplied to the estimator. Write $F_{a,b;u}$ for the u-quantile of an F distribution.

Proposition 6 (Baseline calibration) *Define the matrix-error radius*

$$\widehat{\varepsilon} = \frac{s \sqrt{n^2 F_{n^2, nN; 1-\delta}}}{\sqrt{2 \lambda_{\min}(G)}} + \ell \sqrt{\mathrm{Tr}(G)/\lambda_{\min}(G)}. \tag{13}$$

With probability at least $1 - \delta$, $\|\widehat{M} - M\|_2 \le \widehat{\varepsilon}$. Consequently

$$\widehat{\kappa} = \|\widehat{M}\|_2 + \widehat{\varepsilon} + \ell \tag{14}$$

bounds global baseline sensitivity. Either this scalar bound or the finite-power certificate with $L_\Delta = \widehat{\varepsilon} + \ell$ can supply support. Fresh validation at level α gives unconditional false adoption at most $\alpha + \delta$.

The certificate accounts separately for regression noise, residual bias and the residual's own sensitivity. Appendix A.1 derives the exact pivot. The residual bound

and the action and reward constants remain model assumptions; finite reset data do not establish global regularity. Reset states must also excite every state direction, as expressed by nonsingularity of G.

One freezes the update, calibrates the baseline and tests fresh paired returns. Without a useful certificate, retain the baseline or use a separately justified envelope. Support calibration leaves the sign of the gain to return validation.

4.2 Total cost without one-step contraction

A smaller correction could be costly if it required a progressively more accurate matrix estimate. A fixed positive robust margin avoids this: one fixed accuracy suffices. The next theorem quantifies the acquisition and validation budgets. Fix a block cap K and target matrix accuracy $e_* > 2\ell\sqrt{n}$. For $a_j = \|M^j\|_2$, define deterministic perturbation bounds

$$q_0 = 1, \qquad q_j = a_j + e_* \sum_{s=0}^{j-1} a_{j-1-s} q_s,$$
$$S_* = \sum_{j=0}^{K-1} q_j, \qquad \eta_* = 1 - q_K - (\ell + e_*) S_*. \tag{15}$$

These are analysis constants; the verifier uses only $(\widehat{M}, \widehat{\varepsilon})$ in (10). A positive η_* guarantees that the block certificate remains useful when both matrix error and its reported radius are at most e_*.

Theorem 7 (Total cost) *Assume Gaussian resets $N(0, \sigma_x^2 I_n)$ and $\sigma_x, \sigma > 0$. Fix an integer $K \geq 1$ and $e_* > 2\ell\sqrt{n}$ with $\eta_* > 0$ in (15). Let $\alpha, \beta, \delta \in (0, 1/2)$. Choose N_c by (A.1), calibrate the matrix radius at level $\delta/4$, and search blocks $1, \ldots, K$ as in (10). Retain B if none is certified. For a frozen update satisfying (5) and $\mu_d \geq md$ with fixed $m > 0$, put*

$$C_H = H_\gamma [L_u + (L_x + L_u L_B) L_a S_* / \eta_*]. \tag{16}$$

Under the pointwise reward identity use

$$C_H = H_\gamma L_{r_B} L_a S_* / \eta_*.$$

Let N_v be the ceiling of (3) with $C = C_H$. Apply (2) with Cd replaced by the calibrated b_H from (12). False adoption is at most $\alpha + \delta/4$, power is at least $1 - \beta - \delta$, and the transition cost is at most

$$2N_c + 2HN_v, \tag{17}$$

with N_c, N_v independent of d. When $\gamma < 1$, replacing H_γ by $(1-\gamma)^{-1}$ makes N_v uniform over finite horizons with the same gain margin. For the same favorable law, any ordinary-range rule with power $1 - \beta$ obeys

$$N \geq \frac{B_g}{C_H d} \log \frac{1 - \beta}{\alpha}. \tag{18}$$

Why acquisition remains finite. The explicit allocation in Appendix A.2 ensures, with probability at least $1 - \delta$, that $\|\widehat{M} - M\|_2 \leq \widehat{\varepsilon} \leq e_*$. The perturbation recursion then bounds each p_j by q_j, so $\widehat{\eta}_K \geq \eta_*$ and $\widehat{\Gamma} \leq S_*/\eta_*$. Corollary 5 supplies a support no larger than $C_H d$. The matrix accuracy target is fixed as d shrinks, while support and gain shrink together in validation. The proof and constants are in Appendix A.2.

For a linear baseline ($\ell = 0$), every Schur-stable M has a finite K with $a_K < 1$. Continuity of (15) at $e_* = 0$ then gives a positive admissible target, even if $\|M\|_2 > 1$. For nonlinear residuals, the stated robust margin must still hold. Transient growth now affects the required accuracy and the coefficient S_*/η_* instead of being excluded by one-step contraction.

The allocation assumes bounds on noise, matrix geometry and the gain margin; it is a sufficient budget, not an adaptive sample-selection guarantee. For matched overall power, replace β by $\beta + \delta$ in the ordinary-range comparison. The sequential bound in Appendix B applies with $C = C_H$. An unknown Gaussian covariance can provide the same block gate through Appendix A.4; the explicit allocation here is for isotropic noise.

4.3 A baseline that expands in one step

Consider

$$M = \begin{pmatrix} 1/2 & 1 \\ 0 & 1/2 \end{pmatrix},$$
$$f(x, a) = Mx + .001 \tanh(x) + (0, a)^\top. \tag{19}$$

Use $B = 0$, $\pi_d = d$, $X_0 = 0$, shared i.i.d. disturbances $N(0, .005^2 I_2)$ and reward $r(x, a) = \tanh(x_1)$. Here $\rho(M) = .5$ but $\|M\|_2 = 1.2071$, so the Euclidean one-step criterion fails. With $K = 4$, $e_* = .02$, $\ell = .001$, (15) gives $S_* = 4.1781$ and $\eta_* = .3196$. For $\delta = .05$ and $\sigma_x = 1$, (A.1) requires $N_c = 173$ replicated resets, or 346 calibration transitions. For $0 < d \leq .1$ and $H \geq 3$, the gain has a uniform positive linear lower bound at any fixed $0 < \gamma < 1$; the short proof is in Appendix A.3. Thus the acquisition and pair-budget conclusions apply to an actual stochastic control family outside one-step contraction. This is an analytic example, separate from the frozen numerical study below.

5 Regional support: the price of escape information

The global acquisition result raises a natural qualification: can one obtain the same saving when regularity is known only in a region visited by the baseline? A trajectory tube transfers that regularity to the candidate while the baseline stays in the region. Outside it, the ordinary range must still be charged. The resulting correction depends on escape probability, not only on the local sensitivity coefficient.

Global sensitivity is sufficient but can be unrealistic. Fix a closed region K, baseline, environment, initial-state law and horizon before escape calibration. In the same norm let $K_\rho = \{x : \mathrm{dist}(x, K) \le \rho\}$ and assume the sensitivity and reward bounds of Theorem 2 only on K_ρ. Require $\rho \ge \max_{t \le H} E_t$ and a global return bound $|D| \le B_g$, including trajectories that escape. Define $\mathcal{A} = \{X_t^B \in K,\ 0 \le t \le H\}$ and $p_e = P(\mathcal{A}^c)$.

Proposition 8 (Corrected regional adoption) *Let $b = \min(B_g, b_d) > 0$ (or use b_d^{eq}) and $Y = \mathrm{clip}(D, -b, b)$. An upper confidence bound $\bar{p}_e$ from independent baseline rollouts with error δ_e gives the allowance $q = (B_g - b)\bar{p}_e$. Conditional on its coverage,*

$$|\mathbb{E}Y - \mu| \le q. \tag{20}$$

Betting with factors $1 + \lambda_i(Y_i - q)/(b + q)$, for predictable $0 \le \lambda_i \le 1$, controls false adoption by $\alpha + \delta_e$ at threshold $1/\alpha$. Independently calibrated regional sensitivity adds its own failure probability.

PROOF. On $\mathcal{A}$, the one-step induction in Theorem 2 keeps the repaired state within K_ρ, so $|D| \le b$ and $Y = D$. Everywhere $|Y - D| \le B_g - b$, yielding (20). Under $\mu \le 0$, $\mathbb{E}(Y - q) \le 0$ and $(Y - q)/(b + q) \in [-1, 1]$. The nonnegative supermartingale argument proves the claim; add the escape-calibration failure probability.

Clipping can reduce the gain by q, and correcting the null costs another q: $\mathbb{E}Y - q \ge \mu - 2q$. If $\mu > 2q$, a fixed-sample test $\overline{Y}_N > q + b\sqrt{2\log(1/\alpha)/N}$ has conditional power at least $1 - \beta$ for

$$N \ge \frac{2b^2}{(\mu - 2q)^2}\left(\sqrt{\log(1/\alpha)} + \sqrt{\log(1/\beta)}\right)^2. \tag{21}$$

Thus $b = O(d)$, $\mu \ge md$ and $q \le \eta md$, $0 < \eta < 1/2$, preserve constant validation cost. The remaining question is how much simulation is needed to make q this small.

With zero escapes in N_e independent episodes, the one-sided exact binomial endpoint (Clopper and Pearson, 1934) is

$$\bar{p}_e = 1 - \delta_e^{1/N_e} \le \log(1/\delta_e)/N_e.$$

A sufficient zero-escape-record allocation is $N_e \ge (B_g - b)\log(1/\delta_e)/(\eta md)$, of order d^{-1} for fixed B_g. For this endpoint that order is also required as $d \downarrow 0$, since $1 - e^{-a/N_e} \sim a/N_e$. This cost concerns the stated endpoint on a zero-escape record. Structural information about escape can change it, just as return structure changes the ordinary-range barrier. Each escape episode costs H transitions. Analytic invariance instead sets $p_e = q = 0$ without sampling. A zero observed escape count does not establish invariance, and fixed positive escape probability can dominate arbitrarily small gains even when it is known exactly.

6 Numerical evidence on frozen controller updates

The experiments hold each learned controller fixed and change the noise coupling or support information supplied to its verifier. This separates variance reduction from the saving due to a justified support bound. Training, calibration, certification and independent performance evaluation use separate stages. Plant seeds are the statistical units; means include censored and futility stops. All results below use the previously frozen policies, records and allocations.

6.1 Controlled stochastic plants and protocol

There are 24 plants in dimensions $n = 4, 8, 16$, eight seeds per dimension. Let O, C, U be orthogonal and S skew-symmetric with $\|S\|_2 = 1$. Set $A = .95I + .005S$, $G_0 = .1O$, $K = -.5O^\top$, and target actuator $G_a = G_0 \exp(T)\,\mathrm{diag}(e)$, where T is skew-symmetric, $\|T\|_2 = .35$ and $e_j \in [.8, 1]$. With $Q = U\,\mathrm{diag}(q)U^\top$ and logarithmically spaced $q_j \in [.25, 2]$, use

$$\begin{aligned} X_{t+1} &= AX_t + G_a A_t + .005\tanh(CX_t) + .035Z_t, \\ B(x) &= Kx, \\ r(x, a) &= -.05[1 - e^{-x^\top Qx} + .1(1 - e^{-\|a\|^2})]. \end{aligned} \tag{22}$$

Here $Z_t \sim N(0, I)$, $X_0 \sim N(0, .25I)$, $H = 40$ and $\gamma = e^{-.01}$. Rewards are globally bounded despite unbounded states. The public uncertainty set gives

$$\begin{aligned} \kappa &= \|A + G_0K\|_2 + \tfrac{1}{2}\Delta_G + .005 < 1, \\ \Delta_G &= .1[2\sin(.35/2) + .2]. \end{aligned}$$

Also $L_a = .1$, $L_B = .5$, $L_x = .05\sqrt{4/e}$, $L_u = .005\sqrt{2/e}$ and $L_{r_B} \le L_x + L_u L_B$. The ordinary half-width is $B_g = .055\sum_{t<40}\gamma^t$.

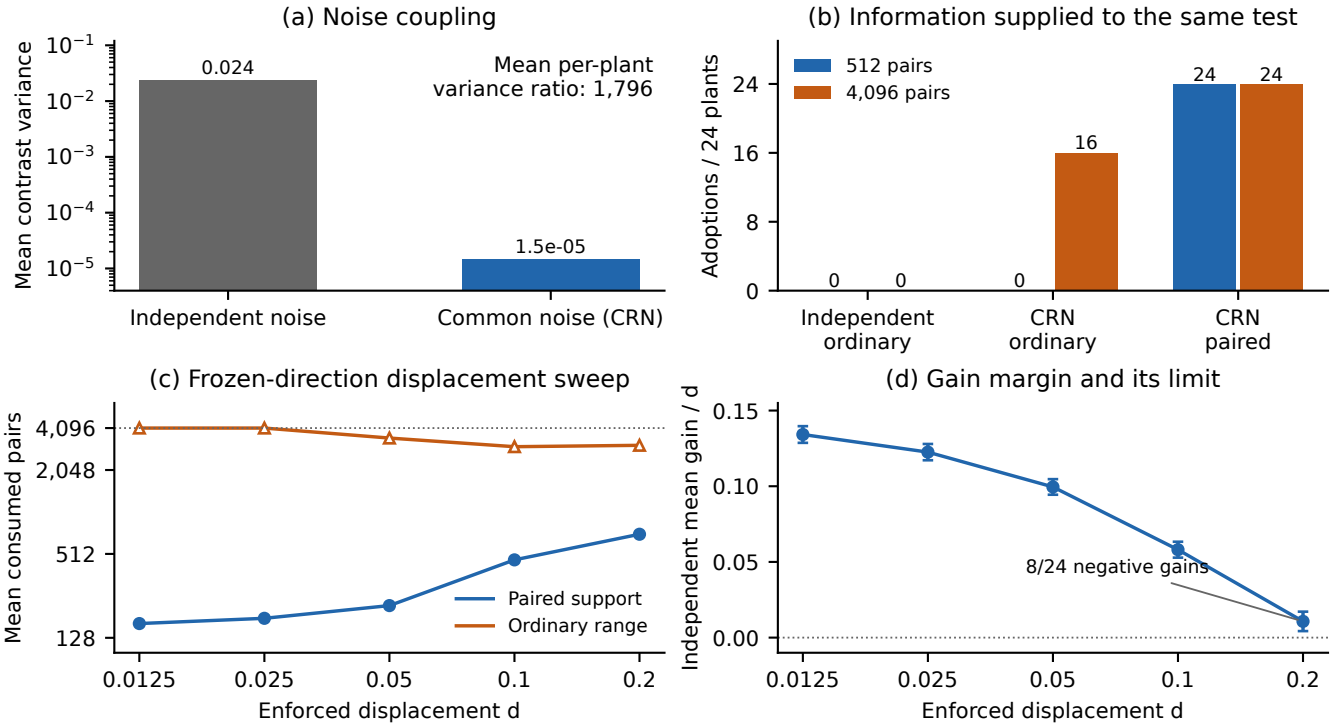


Fig. 1. The same 24 controlled plants. (a) Independent-noise and CRN contrast variances; the mean per-plant ratio is 1,796. (b) Adoptions under ordinary range and paired support. (c) Mean pairs in the fresh-noise displacement sweep, including futility and censoring at 4,096 (open markers). (d) Independently evaluated gain/displacement, with across-plant SE. Directions are rescaled without retraining; panels are related comparisons.

A ridge learner scores rotations of $B(x)$ using paired-return queries, preserves the baseline action norm and enforces $d = .05$. It receives no target matrices or simulator derivatives and retains the unchanged action when no score is positive. Query branches share noise and follow the baseline after the queried action. Anchor and query trajectories together cost 81,920 training transitions. The artifact specifies features, rotation limits, ridge penalty, query allocations and seeds. The resulting policy can be discontinuous.

All fixed-policy comparisons use the following validation protocol. The fixed mixture (4) has stakes $1, 1/2, \ldots, 1/128$ and weights $1/2, 1/14, \ldots, 1/14$. Inspect every 16 pairs and at the cap; a separate negative-tail test can stop for futility. Each pair costs 80 transitions. The ordinary and paired-support tests receive the same ordered differences, except in the explicit independent-noise ablation. Independent evaluation uses 4,096 pairs. Known-support comparisons retain $\alpha = .05/(24 \cdot 2 \cdot 3 \cdot 2 \cdot 2)$; calibrated comparisons retain $\delta_c = .025/(24 \cdot 3)$ and $\alpha = .025/(24 \cdot 3 \cdot 2 \cdot 3 \cdot 2 \cdot 2)$. These are the original family allocations, not retuned levels.

6.2 Variance, support and charged calibration

At $d = .05$, CRN reduces contrast variance by a mean per-plant factor of 1,796 (range 933–2,774). Neither independent nor CRN evaluation with ordinary range adopts at 512 pairs. With CRN and paired support the same fixed mixture adopts all 24 (Fig. 1). At 4,096 pairs, CRN/ordinary range adopts 16/24 versus zero with independent noise; variance reduction helps, but does not explain the primary-budget saving.

Information	Adopt	Deployed gain ($\times 10^3$)	Transitions used
Known support	24/24	$4.995 \pm .255$	99,947
Ordinary range	0/24	0	122,880
Calibrated support	22/24	$4.729 \pm .375$	110,496
Ordinary (+pairs)	0/24	0	126,960

Table 1
Primary frozen-policy comparisons. Deployed gain is measured on the independent evaluation bank; $\pm$ is across-plant SE. Mean transitions used include training, calibration and validation until stopping. Known-support arms allow 512 pairs. Calibration costs 4,096 transitions and retains 512 pairs; its ordinary comparator instead receives 563 pairs. Allocated totals are 122,880 and 126,976; the latter ordinary arm leaves 16 transitions unused.

With known residual limit $\ell = .005$, 2,048 replicated resets cost 4,096 transitions. Charging this acquisition cost still permits 22/24 adoptions versus 0/24 when the or-

dinary comparator spends the same allowance on extra pairs (Table 1). Known-support and calibrated comparisons reuse policies and return banks; their outcomes are not independent replications. Rescaling the 24 frozen directions from $d = .05$ to .0125 preserves positive independently estimated gains and mean gain/d of about .100–.134. Supported betting uses 162–218 mean pairs and adopts all updates. Both smallest-displacement ordinary-range conditions are censored at 4,096, so the experiment does not estimate their asymptotic d^{-1} rate. At $d = .2$, mean gain/d falls to .011; eight estimated gains become negative, and supported betting retains the baseline on those eight. The gain-margin condition cannot be extrapolated to large updates.

6.3 An underactuated mechanical example

The scalar plants establish the coupling and support mechanism. We next ask whether retaining known actuator directions makes a practical difference when matrix propagation is used. This comparison changes the envelope while keeping the matrix-confidence event, learned update and return observations fixed. On 16 separately frozen underactuated spring–mass plants, retaining $\widehat{M}^j G_0$ in Proposition 4 permits 16/16 primary adoptions, versus none for either the unrefined matrix envelope or ordinary range. For $p = 4, 8$ positions and p velocities, the nominal Euler matrices are

$$A_0 = \begin{pmatrix} I & .05I \\ -.05S_m & I - .05D_m \end{pmatrix}, \quad G_0 = \begin{pmatrix} 0 \\ .05E \end{pmatrix}.$$

Here $S_m = \mathrm{diag}(s) + .25L^\top L$, $s_j \in [.8, 1.2]$, $L = I - I_{+1}$, D_m has diagonal entries in $[.35, .45]$, and E selects alternating force inputs. Nominal LQR weights $Q_m = \mathrm{diag}(2I, .3I)$ and $R_m = .5I$ define the baseline. The target uses $.85D_m$, an actuator rotation of spectral angle .35, efficiencies in $[.8, 1]$, and velocity residual $-.001\tanh(q)$. The residual limit is .001, displacement is .15, and the reward uses Q_m in (22) with effort weight .02. Reset covariance is $.09I$, and the unknown disturbance covariance has rotated eigenvalues between $.5(.015)^2$ and $2(.015)^2$.

There are $p/2$ force inputs, so corrections enter only selected velocity coordinates. The position–velocity coupling then spreads their influence through the baseline dynamics. The products $\widehat{M}^j G_0$ retain this route of entry; replacing them by $\|\widehat{M}^j\|_2\|G_0\|_2$ also allows forcing in directions the actuator cannot reach in one step. Calibration uses 8,192 replicated resets and the rowwise covariance pivot (A.3); supported tests have 512 pairs and the equal-cost ordinary arm has 716. At an extended common cap of 4,096, all three tests succeed, with mean counts 295, 587 and 2,582 respectively. Table 2 separates the primary budget from the extended common budget.

Envelope	Primary adoptions	Extended mean pairs
Actuator-aware matrix	16/16	295
Unrefined matrix	0/16	587
Ordinary range	0/16	2,582

Table 2
The same 16 frozen mechanical updates. Supported primary tests allow 512 pairs; the ordinary comparator allows 716 after its calibration saving is converted to return pairs. Extended tests all allow 4,096 pairs and all adopt 16/16. Counts are rounded means.

d	Ordinary	Finite-H	Block	Model
.05	1,824	442	1,018	346
.025	3,648	442	1,018	346
.0125	7,200	442	1,018	346

Table 3
Mean transitions consumed until decision; all methods adopt 24/24 positive updates. Informed methods include 346 calibration transitions. Finite-H: Proposition 4; Block: Corollary 5; Model: Appendix C, using the same calibration data without return validation.

The unrefined envelope eventually succeeds too; the observed benefit is that the actuator-aware bound makes the same calibrated information useful with fewer return pairs.

The frozen dynamics, policies, banks and allocations are included in the artifact. This experiment illustrates how actuator geometry can make a calibrated support useful at a limited validation budget.

6.4 Finite-power acquisition and a model-based reference

We additionally test (19) at $H = 3$ and $\gamma = .95$, with 24 independent calibration/validation replications. Each uses 173 replicated resets: no one-step contraction gate passes, but all certify block length four. Methods share banks and the known actuator image. Each decision has error allowance .05: matrix coverage uses .0125 and calibrated return testing uses .0375. The ordinary test uses .05 and $B_g = 2(\gamma + \gamma^2)$, since the initial reward is zero. Tests use (4), inspected every 16 pairs, with caps 2,048 (supported) and 2,105 (ordinary, spending its calibration saving). Displacements reuse noises. Zero and negative controls produce no adoptions.

Block validation uses 112 pairs at each displacement, versus 16 for the tighter finite-horizon envelope. The model bound is already positive from calibration: control information can support direct gain certification too. The block bound is uniform in time; the model check exploits the special horizon-three structure.

7 Scope of the certification claim

The gain margin $\mu_d \geq md$ is a condition for verification power, not a guarantee from learning. The block theorem needs a positive robust margin. Section 6.4 checks its calibrated support with sequential testing; it does not assess tightness of the fixed-sample sufficient allocation in Theorem 7.

Residual, action and reward regularity remain model assumptions. Replicated transitions certify matrix uncertainty, not global regularity. Regional information requires the escape correction in Section 5 or a valid invariance argument. Expected-return improvement does not establish state-constraint safety or transfer automatically to a different physical plant.

8 Conclusion

Baseline sensitivity links controller displacement to paired-return support. For proportional gains and a fixed robust margin, finite-power calibration and validation have displacement-independent budgets, including transiently expanding baselines. Under fixed discount the pair budget is uniform in finite horizon. Regional escape information can restore inverse-displacement cost. The resulting design question is which information suffices for a reliable adoption decision and whether its acquisition costs less than the validation it saves.

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The authors declare no competing interests.

CRediT authorship contribution statement

Kyunghyun Lee: Methodology, software, investigation, validation, visualization, writing—original draft, writing—review and editing. **Yeoneung Kim:** Conceptualization, methodology, formal analysis, supervision, writing—original draft, writing—review and editing.

Data availability

Frozen experiment code and records are available at https://github.com/yeoneung/crn_policy. The additional finite-power study code and records accompany the submission package. The manuscript uses synthetic data only.

Declaration of generative AI use

GPT (OpenAI) assisted research and manuscript preparation. The authors reviewed and edited the output and take full responsibility for the manuscript.

A Calibration calculations

A.1 Proof of Proposition 6

PROOF. Conditional on X, write H_X for the matrix of residual values and $\overline{W} = (W_1 + W_2)/2$. Regression gives

$$\widehat{M} - M = (X^+ H_X)^\top + (X^+ \overline{W})^\top, \qquad X^+ = G^{-1} X^\top.$$

The residual bias is at most $\ell\sqrt{\mathrm{Tr}(G)/\lambda_{\min}(G)}$, since $\|H_X\|_F^2 \leq \ell^2 \,\mathrm{Tr}(G)$. The squared Frobenius norm of $G^{-1/2} X^\top \overline{W}$, divided by $\sigma^2/2$, has law $\chi^2_{n^2}$. Independently, $nNs^2/\sigma^2 \sim \chi^2_{nN}$, because the replicate difference cancels the common residual and is independent of the replicate mean. The F pivot bounds regression noise by the first term in (13); the bias bound gives the second term. Finally $\mathrm{Lip}(F_B) \leq \|M\|_2 + \ell$. Conditional on valid calibration, either scalar or block support gives validation error at most α; add the calibration failure δ.

A.2 Allocation and proof of Theorem 7

Set $L = \log(8/\delta)$ and $t = \sqrt{2\log(4/\delta)}$. It suffices to take

$$N_c = \left\lceil \max\left\{ \left[\frac{12\sigma(n + \sqrt{2L})}{5\sigma_x(e_* - 2\ell\sqrt{n})} \right]^2, \; 9(\sqrt{n} + t)^2, \; 16L/n \right\} \right\rceil. \tag{A.1}$$

PROOF. Gaussian singular-value bounds (Vershynin, 2012, Corollary 5.35) give

$$\lambda_{\min}(G) \geq \tfrac{4}{9}\sigma_x^2 N_c, \qquad \mathrm{Tr}(G)/\lambda_{\min}(G) \leq 4n$$

except with probability $\delta/2$. The chi-square bounds (Laurent and Massart, 2000, Lemma 1), using $nN_c \geq 16L$, give $s^2 \leq 2.5\sigma^2$ except with probability at most $\delta/4$. For independent $U \sim \chi^2_{n^2}$ and $V \sim \chi^2_{nN_c}$,

$$P(V/(nN_c) < 1/2) \leq e^{-nN_c/16} \leq \delta/8$$

and $P(U > (n + \sqrt{2L})^2) \leq \delta/8$. Hence

$$\sqrt{n^2 F_{n^2, nN_c; 1-\delta/4}} \leq \sqrt{2}(n + \sqrt{2L}).$$

On the design and scale events, the reported radius at level $\delta/4$ therefore satisfies

$$\widehat{\varepsilon} \le 2\ell\sqrt{n} + \frac{12\sigma(n+\sqrt{2L})}{5\sigma_x\sqrt{N_c}} \le e_*;$$

here $\frac{3}{2}\sqrt{2.5} < 12/5$. Intersecting with matrix coverage gives total failure probability at most δ.

Let $\Delta = \widehat{M} - M$. The identity

$$\widehat{M}^j - M^j = \sum_{s=0}^{j-1} M^{j-1-s}\Delta\widehat{M}^s$$

and induction give $p_j \le q_j$ when $\|\Delta\|_2 \le e_*$. Consequently $\widehat{S}_K \le S_*$, $\widehat{\eta}_K \ge \eta_* > 0$, and minimizing over certified blocks gives $\widehat{\Gamma} \le S_*/\eta_*$. Fresh return validation then has conditional power $1-\beta$ with

$$N_v = \left\lceil 2(C_H/m)^2\left(\sqrt{\log(1/\alpha)} + \sqrt{\log(1/\beta)}\right)^2 \right\rceil. \quad \text{(A.2)}$$

Adding the useful-calibration failure gives power $1-\beta-\delta$. Matrix coverage alone makes every selected block valid, so false adoption is at most $\alpha + \delta/4$, including cases in which the reported support is conservative. The two transition counts give (17). Applying Corollary 5 to the true M with zero matrix error also gives $\mu_d \le C_H d$; Proposition 1 and $\log(1+x) \le x$ prove (18). The horizon statement follows from $H_\gamma \le (1-\gamma)^{-1}$.

The one-step condition is recovered by $K = 1$: if $g = 1 - \|M\|_2 - \ell(1+4\sqrt{n}) > 0$, choose $e_* = 2\ell\sqrt{n} + g/4$ to obtain $\eta_* = g/2$ and $S_* = 1$.

A.3 Gain in the nonnormal example

In (19), the shared-noise state difference is componentwise nonnegative: M has nonnegative entries, tanh is increasing, and the update adds d in the second coordinate. Its first coordinate equals d at time two. All stage-return differences are therefore nonnegative, and, on $|X^B_{2,1}| \le 1$, the time-two difference is at least $d\,\mathrm{sech}^2(1.1)$ for $d \le .1$. Write $\sigma = .005$. The baseline coordinate satisfies $X^B_{2,1} = Z + .001\tanh(W_{0,1})$ with $Z \sim N(0,(1.5\sigma)^2)$. Since the residual magnitude is at most .001, $P(|X^B_{2,1}| \le 1) \ge p_0$, where Φ is the standard normal CDF and

$$p_0 = 2\Phi\left(\frac{1-.001}{1.5\sigma}\right) - 1 > 0,$$
$$\mu_{d,H} \ge \gamma^2 p_0\,\mathrm{sech}^2(1.1)d$$

for every $H \ge 3$. Corollary 5 gives the matching $O(d)$ upper bound, uniform in finite horizon for fixed $\gamma < 1$.

A.4 Unknown Gaussian covariance

Unknown positive-definite Gaussian noise covariance is compatible with the matrix envelope. Set $s_j^2 = \|(Y_1 - Y_2)_{:j}\|^2/(2N)$ and $c = \sqrt{nF_{n,N;1-\delta/n}}$. A valid matrix-error bound is

$$\varepsilon = \frac{c\sqrt{\sum_j s_j^2/2}}{\sqrt{\lambda_{\min}(G)}} + \ell\sqrt{\mathrm{Tr}(G)/\lambda_{\min}(G)}. \quad \text{(A.3)}$$

Write the covariance as $\Sigma \succ 0$. Conditional on the resets,

$$\frac{\|G^{-1/2}X^\top \overline{W}_{:j}\|^2}{\Sigma_{jj}/2} \sim \chi^2_n, \qquad \frac{Ns_j^2}{\Sigma_{jj}} \sim \chi^2_N.$$

The two variables are independent because the full Gaussian replicate mean and difference are independent. Dividing their normalized versions gives the coordinate pivot. On its coverage event the squared norm of that regression-error row is at most $c^2 s_j^2/(2\lambda_{\min}(G))$. Intersecting the n coordinate events gives coverage at least $1-\delta$, and summing these squared row bounds yields the noise term in (A.3). Cross-coordinate correlation changes neither this marginal argument nor the union bound. This extension supplies a matrix-confidence event for the actuator envelope and finite-power gate; the explicit allocation in Theorem 7 remains the isotropic-noise statement.

B Power of a specified betting component

On the useful calibration event, $Z_i = D_i/b \in [-1,1]$ has mean at least $\nu = m/C_H \le 1$. A component of (4) with stake $\lambda = \nu/4$ and weight $w > 0$, without early futility stopping, crosses with conditional probability $1-\beta$ by

$$N \ge \max\{8\nu^{-2}\log(1/\beta),\ 16\nu^{-2}\log(1/(\alpha w))\}.$$

Indeed Hoeffding gives $\overline{Z}_N \ge \nu/2$ on that event, while $\log(1+x) \ge x - x^2$ for $|x| \le 1/2$ gives $\log M_N \ge \log w + N\nu^2/16$. This power statement concerns the specified stake, not every finite experimental grid.

C Model-based reference for the additional example

At $H = 3$ the first two rewards agree. The time-two state difference in coordinate one lies in $[c_- d, c_+ d]$, where $c_\pm = \widehat{M}_{12} \pm (\widehat{\varepsilon} + \ell)$. The same replicates give $\bar{\sigma}^2 = 2N_c s^2/\chi^2_{2N_c;\delta_s}$, using the lower quantile with $\delta_s = .0375$. With matrix coverage at .0125, total failure is at most .05. Since $h(0) = 0$, the baseline coordinate is $X^B_{2,1} = M_{1,:}W_0 + h_1(W_0) + W_{1,1}$. Minkowski's inequality and independence of the centered W_1 give

$$\mathbb{E}|X^B_{2,1}|^2 \le v := \bar{\sigma}^2[(\|\widehat{M}_{1,:}\|_2 + \widehat{\varepsilon} + \sqrt{2}\ell)^2 + 1].$$

For $c_- > 0$, Markov's inequality and the mean value theorem give

$$\mu_d \geq \gamma^2 c_- d(1-v)_+ \operatorname{sech}^2(1 + c_+ d). \tag{C.1}$$

Indeed $P(|X^B_{2,1}| \leq 1) \geq (1-v)_+$; there the derivative bound applies, and elsewhere the reward difference is nonnegative. A positive bound permits adoption without return validation. Reflected bounds handle negative actions; zero action gives zero gain.

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